Chapter 8: Vector Norms - Summary

Chapter 8 introduces vector norms, which measure the length or magnitude of a vector. Three norms are emphasized: 1. L1 Norm (Manhattan Norm): Sum of the absolute values of the vector components. Useful for sparsity and LASSO-style regularization. 2. L2 Norm (Euclidean Norm): Square root of the sum of squared components. This is the most common notion of vector length and is widely used in machine learning, least-squares regression, optimization, and regularization. 3. Max Norm (Infinity Norm): The largest absolute component in the vector. Used in some optimization and neural-network regularization methods. Key ideas: - Norms quantify vector magnitude. - Different norms emphasize different notions of "size." - Norms are heavily used in machine learning, especially regularization. - L2 norm is the most common norm encountered in regression and statistics.